

Powers of Ten

Every graded set, every week — 4 weeks, 20 two-page sheets, with an answer key after each week.

WEEK 1

5 sets

WEEK 2

5 sets

WEEK 3

5 sets

WEEK 4

5 sets

HOW TO USE IT

One page per day. Warm-Up is recall, Core is the graded tier, Challenge is never graded — the score box counts the graded items only. The same items appear in the app's Practice Bay, so a day can be done on paper or on screen, not both.

Ten Times Bigger

Every place is ten of the place to its right. Say it as you write it.

Warm-Up 12 ITEMS • RECALL

1 The 4 in 40 is worth _____

2 The 4 in 400 is worth _____

3 Ten times 30 _____

4 Ten times 700 _____

5 One tenth of 500 _____

6 One tenth of 60 _____

7 3×10 _____8 3×100 _____9 6×10 _____10 12×10 _____11 9×100 _____12 40×10 _____

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. The 7 in 7,000 is how many times the 7 in 700

14. The 3 in 300 is how many times the 3 in 3 (Two jumps left)

15. 100 times 40

16. One hundredth of 900

17. The 5 in 50,000 is worth

18. How many times bigger is 500 than 50

19. How many times bigger is 4000 than 40

20. The 7 in 7,000 is worth how many times the 7 in 700

21. 2.5×10 22. 0.9×100



Challenge

NOT GRADED • REASON IT OUT

23. The 6 in 60,000 is how many times the 6 in 60 (Three jumps)

(Three jumps)

24. A digit moves two places left. Its value is multiplied by

25. How many times bigger is 6,000 than 0.6

26. The 3 in 3,000 compared with the 3 in 0.3



Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE



Error journal

ONE SENTENCE PER MISTAKE

Down to Thousandths

Three places past the point. Type decimals with a leading zero, like 0.45.

Warm-Up 11 ITEMS • RECALL

1 Write four tenths as a decimal

2 Write seven hundredths

3 Write three thousandths

4 Write twenty-five hundredths

5 The 5 in 0.5 means five

6 Write one half as a decimal

7 Digits after the point in 0.451

8 The place of the 5 in 0.05 — type tenths, hundredths or thousandths

9 0.7 in words — type tenths or hundredths

10 Write nine hundredths

11 Write six thousandths

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

12. Write four hundred six thousandths

13. Write two and thirty-five hundredths

14. 0.60 written more simply

15. Write sixty-two thousandths

16. How many hundredths in 0.3

17. Write three hundred twenty-five thousandths

18. The value of the 8 in 4.081 — type as a decimal

19. $0.4 + 0.06 + 0.003$

20. Write 2 and 15 hundredths

21. 0.60 and 0.6 — type same or different



Working space



Error journal

Powers of Ten

The digits move. The point stays put.

Warm-Up 12 ITEMS • RECALL

1 3.5×10

2 3.5×100

3 $42 \div 10$

4 $42 \div 100$

5 10^2 as a number

6 10^3 as a number

7 10^2

8 10^3

9 10^5

10 Zeros in 10^7

11 10^1

12 10^0

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. 0.06×100

14. $7 \div 1000$

15. 2.4×10^3

16. $850 \div 10^2$

17. 10^4 as a number

18. 5×10^3

19. 8×10^2

20. 1.2×10^3

21. 6.5×10^4

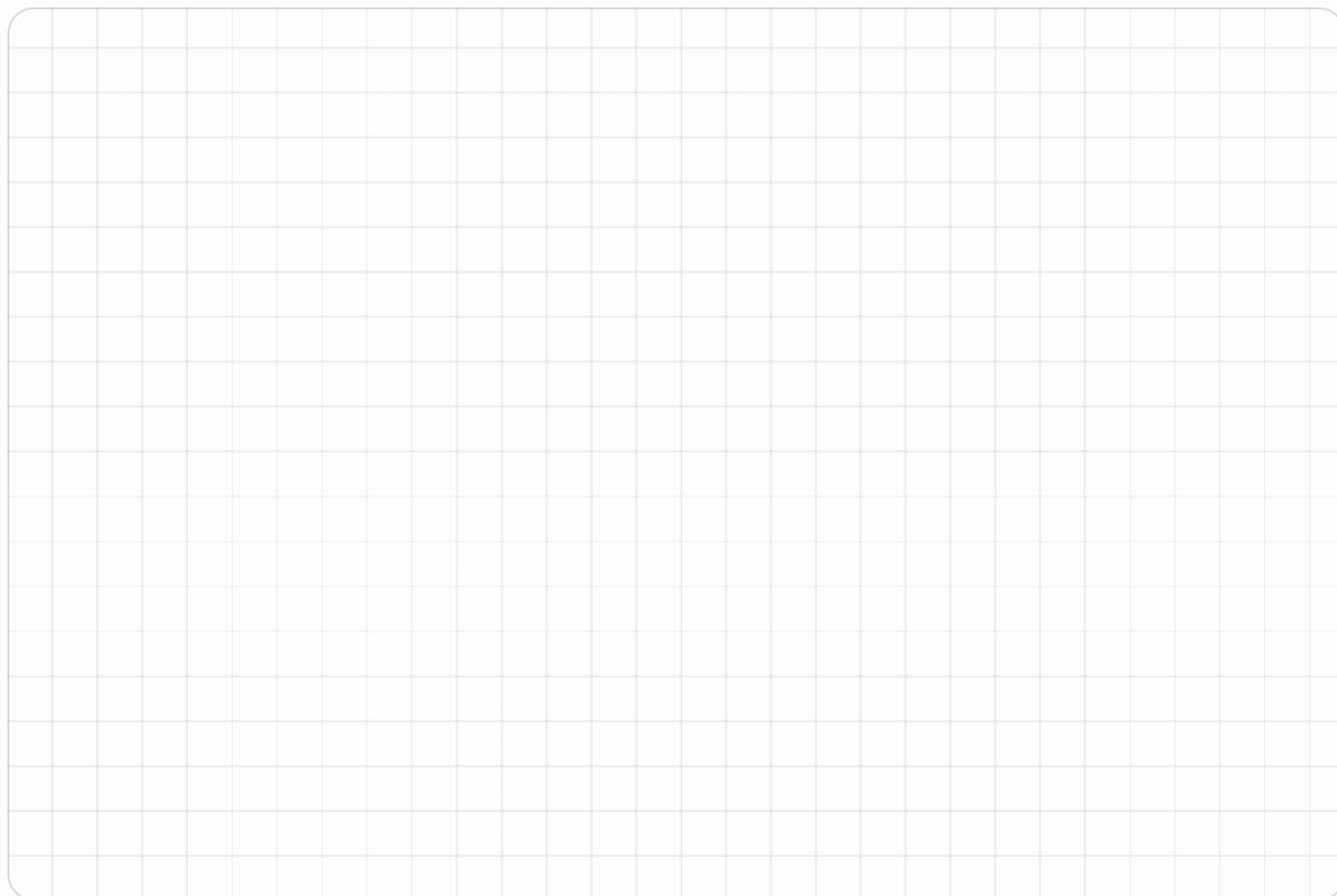
22. 0.4×10^2

★ Challenge NOT GRADED • REASON IT OUT

23. 0.035×10^4
.....

24. Which power of ten takes 0.7 to 700 (Three places)
.....

25. $10^2 \times 10^3$
.....

26. Which exponent takes 3 to 30,000
.....**✎ Working space** SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Compare & Round

Line up the points. Trailing zeros cost nothing.

Warm-Up 12 ITEMS • RECALL

1 Larger: 0.5 or 0.35 — type it

2 Larger: 0.7 or 0.70

3 Round 4.7 to the nearest whole

4 Round 0.34 to the nearest tenth

5 Round 128 to the nearest ten

6 Round 0.86 to the nearest tenth

7 Bigger — type 0.5 or 0.4

8 Bigger — type 0.7 or 0.68

9 Round 4.6 to the nearest whole

10 Round 8.2 to the nearest whole

11 Round 0.45 to the nearest tenth

12 Bigger — type 1.2 or 1.02

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Round 2.451 to the nearest hundredth

14. Round 9.96 to the nearest tenth

15. Larger: 0.406 or 0.41

16. Round 0.0475 to the nearest thousandth

17. Round 4,829 to the nearest hundred

18. Round 3.847 to the nearest hundredth

19. Round 12.451 to the nearest tenth

20. Round 0.0649 to the nearest hundredth

21. Bigger — type 0.309 or 0.31

22. Round 99.7 to the nearest whole



Decimal Duel & Zoom Atlas

Enrichment. Build big, then justify it.

✦ Warm-Up 4 ITEMS • RECALL

1 Biggest number from digits 4, 7, 1, 9

2 Smallest number from those digits

3 $0.3 + 0.4$

4 10×10

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

5. Biggest decimal under 1 from 4, 7, 1 — type it

.....

6. A model is 100 times a 3 cm object. Its size in cm

.....

7. 4.5×100

.....

8. Round 6.482 to the nearest tenth

.....

 Challenge NOT GRADED • REASON IT OUT

9. An object is drawn one thousandth of true size. True length 4000 mm, so drawn length in mm

10. Digits 2, 5, 8: the value closest to 1 you can build as 0.____ (Biggest is nearest to one)

11. How many times bigger is 8.5 than 0.085

12. 0.007×10^4

13. How many times bigger is 90 than 0.09

14. Write five and eight thousandths

 Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal ONE SENTENCE PER MISTAKE

Week 1 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

1.1 • Ten Times Bigger

OUT OF 22

Warm-Up • 1-12 • 40 | 400 | 300 | 7000 | 50 | 6 | 30 | 300 | 60 | 120 | 900 | 400

Core • 13-22 • 10 | 100 | 4000 | 9 | 50000 | 10 | 100 | 10 | 25 | 90

Challenge • 23-26 • 1000 | 100 | 10000 | 10000

1.2 • Down to Thousandths

OUT OF 21

Warm-Up • 1-11 • 0.4 | 0.07 | 0.003 | 0.25 | tenths | 0.5 | 3 | hundredths | tenths | 0.09 | 0.006

Core • 12-21 • 0.406 | 2.35 | 0.6 | 0.062 | 30 | 0.325 | 0.08 | 0.463 | 2.15 | same

Challenge • 22-25 • 9.009 | 250 | 7.042 | 0.09

1.3 • Powers of Ten

OUT OF 22

Warm-Up • 1-12 • 35 | 350 | 4.2 | 0.42 | 100 | 1000 | 100 | 1000 | 100000 | 7 | 10 | 1

Core • 13-22 • 6 | 0.007 | 2400 | 8.5 | 10000 | 5000 | 800 | 1200 | 65000 | 40

Challenge • 23-26 • 350 | 1000 | 100000 | 4

1.4 • Compare & Round

OUT OF 22

Warm-Up • 1-12 • 0.5 | equal | 5 | 0.3 | 130 | 0.9 | 0.5 | 0.7 | 5 | 8 | 0.5 | 1.2

Core • 13-22 • 2.45 | 10 | 0.41 | 0.048 | 4800 | 3.85 | 12.5 | 0.06 | 0.31 | 100

Challenge • 23-26 • 5.25 | 1 | 7.49 | 4.5

Fri • Decimal Duel & Zoom Atlas

OUT OF 8

Warm-Up • 1-4 • 9741 | 1479 | 0.7 | 100

Core • 5-8 • 0.741 | 300 | 450 | 6.5

Challenge • 9-14 • 4 | 0.852 | 100 | 70 | 1000 | 5.008

Multiply by Ten

The digits move. The point stays exactly where it was.

✦ Warm-Up 12 ITEMS • RECALL

1 4.2×10

2 4.2×100

3 0.7×10

4 0.7×100

5 3×10

6 3×1000

7 5.1×10

8 0.3×100

9 8×100

10 1.4×10

11 6×10

12 0.9×10

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. 0.06×100

14. 2.4×1000

15. 0.035×100

16. 12.5×10

17. 0.008×1000

18. 0.07×100

19. 3.6×1000

20. 0.045×100

21. 22.5×10

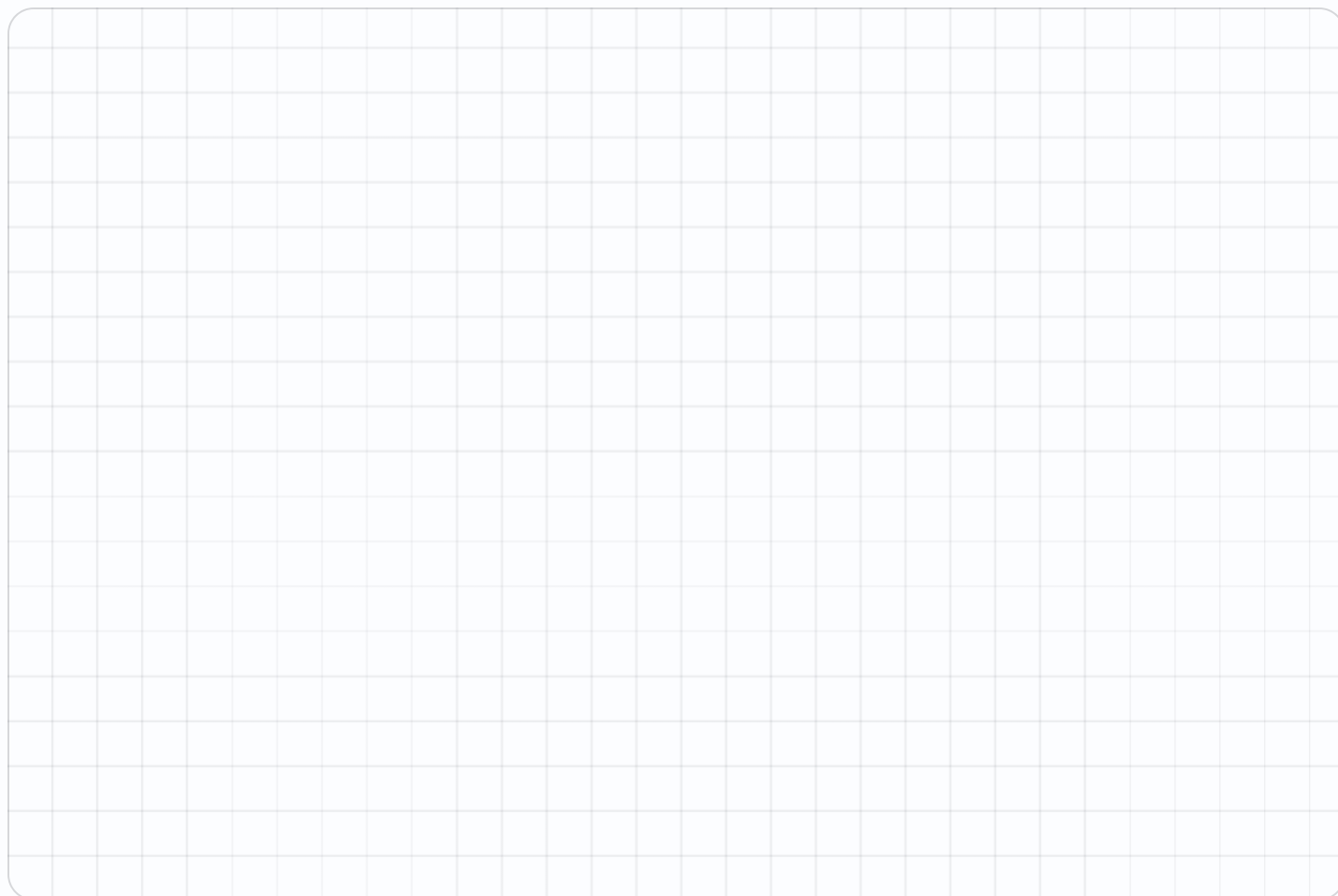
22. 0.006×1000

★ Challenge NOT GRADED • REASON IT OUT

23. 0.0405×1000
.....

24. Which power of ten takes 0.7 to 7000
.....

25. 0.0302×1000
.....

26. Which power of ten takes 0.4 to 4000
.....**✎ Working space** SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Divide by Ten

The same journey in reverse.

Warm-Up 12 ITEMS • RECALL

1 $42 \div 10$

2 $420 \div 100$

3 $7 \div 10$

4 $70 \div 100$

5 $300 \div 10$

6 $5000 \div 1000$

7 $51 \div 10$

8 $300 \div 100$

9 $8 \div 10$

10 $90 \div 100$

11 $600 \div 10$

12 $7000 \div 1000$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $6 \div 100$

14. $850 \div 100$

15. $7 \div 1000$

16. $125 \div 10$

17. $9 \div 10$

18. $4 \div 100$

19. $920 \div 100$

20. $3 \div 1000$

21. $225 \div 10$

22. $6 \div 10$



Exponent Shorthand

10^3 means three tens multiplied, which is three places.

Warm-Up 8 ITEMS • RECALL

1 10^1 2 10^2 3 10^3 4 10^4 5 Zeros in 10^5 6 10^6 7 Zeros in 10^6 8 10^5

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

9. 3×10^2 10. 4.5×10^3 11. 7×10^4 12. 0.6×10^3 13. 2.5×10^2 14. 6×10^2 15. 3.5×10^3 16. 9×10^4 17. 0.8×10^3 18. 1.5×10^2

★ Challenge NOT GRADED • REASON IT OUT

19. $10^3 \times 10^2$

.....

20. 0.035×10^4

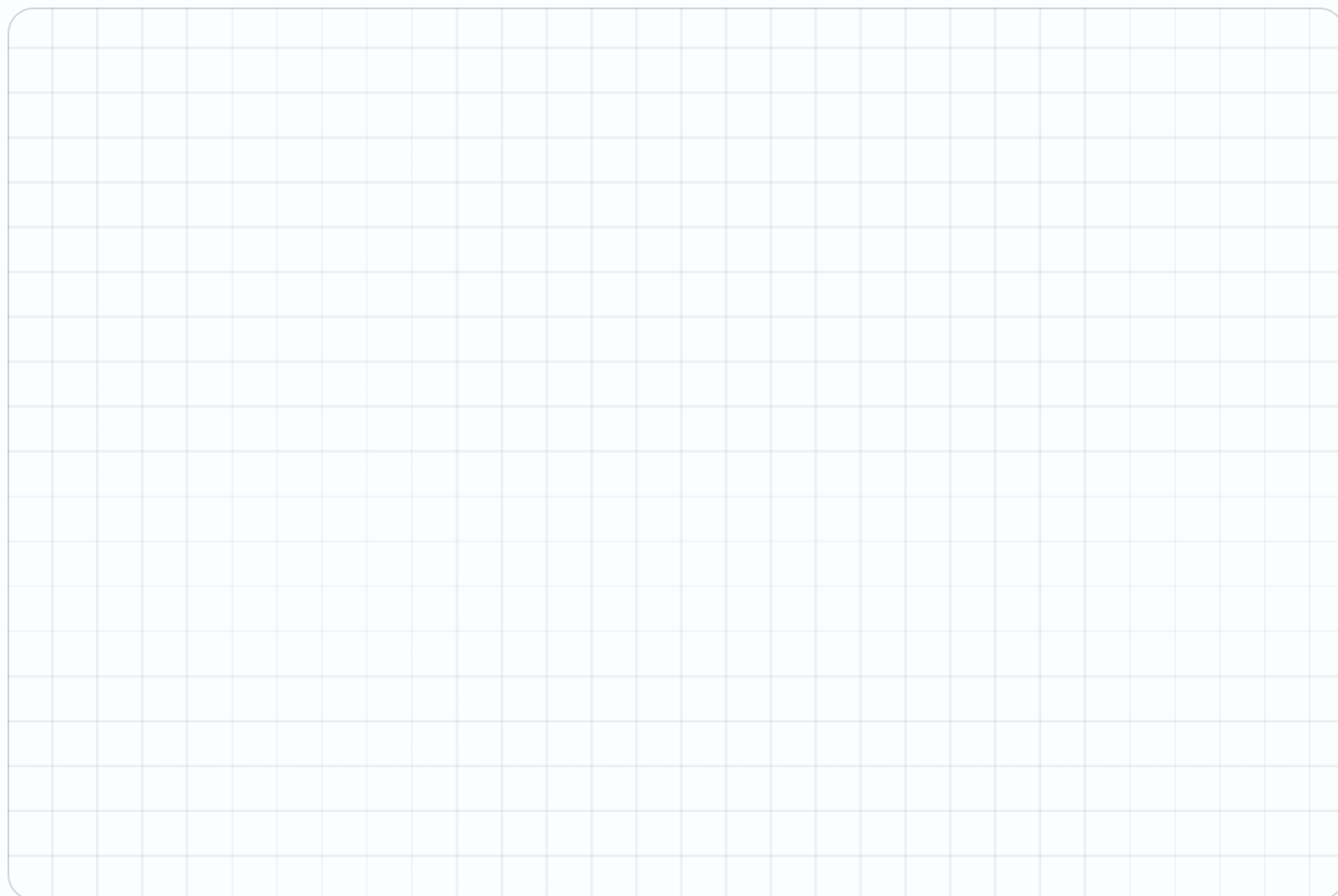
.....

21. $10^4 \times 10^2$

.....

22. 0.045×10^4

.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Patterns in Zeros

Why 4×100 ends in two zeros and 0.4×100 does not.

Warm-Up 12 ITEMS • RECALL

1 4×100

2 0.4×100

3 40×100

4 0.04×100

5 Zeros at the end of 4×1000

6 0.4×1000

7 6×100

8 0.6×100

9 60×100

10 0.06×100

11 0.6×1000

12 Zeros at the end of 6×1000

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. 25×100

14. 0.25×100

15. 0.025×100

16. 1.5×1000

17. 0.015×1000

18. 35×100

19. 0.35×100

20. 0.035×100

21. 2.5×1000

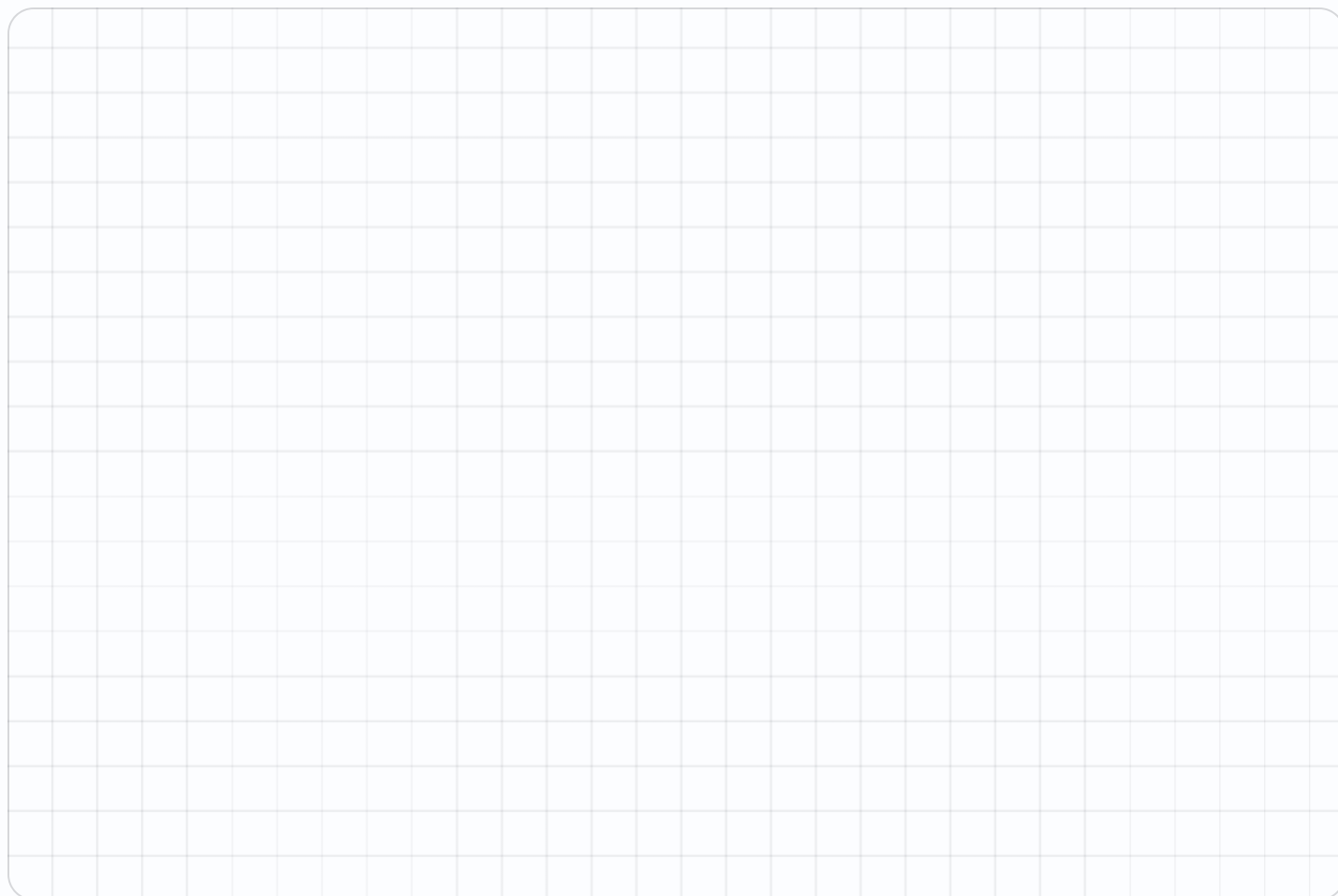
22. 0.025×1000

★ Challenge NOT GRADED • REASON IT OUT

23. 0.0025×10000
.....

24. How many times bigger is 8.5 than 0.085
.....

25. 0.0045×10000
.....

26. How many times bigger is 9.2 than 0.092
.....**✎ Working space** SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Zoom Atlas Begins

Pick the object. Draw it at true size first.

✦ Warm-Up 3 ITEMS • RECALL

1 A 3 cm object at 10 times
— cm

2 At 100 times — cm

3 A 5 cm object at 10 times
— cm

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

4. A 4 m object at one hundredth — cm

.....

5. A 250 cm object at one tenth — cm

.....

6. A 6 m object at one hundredth — cm

.....

7. A 480 cm object at one tenth — cm

.....

★

Challenge

NOT GRADED · REASON IT OUT

8. A 4000 mm object at one thousandth — mm
-
9. A 2.5 cm object at 1000 times — metres
-
10. From 0.5 mm to 5 m — the multiplier
-
11. A 7000 mm object at one thousandth — mm
-
12. A 3.5 cm object at 1000 times — metres
-
13. From 0.2 mm to 2 m — the multiplier
-

Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

!

Error journal

ONE SENTENCE PER MISTAKE

Week 2 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

2.1 • Multiply by Ten

OUT OF 22

Warm-Up • 1-12 • 42 | 420 | 7 | 70 | 30 | 3000 | 51 | 30 | 800 | 14 | 60 | 9

Core • 13-22 • 6 | 2400 | 3.5 | 125 | 8 | 7 | 3600 | 4.5 | 225 | 6

Challenge • 23-26 • 40.5 | 10000 | 30.2 | 10000

2.2 • Divide by Ten

OUT OF 22

Warm-Up • 1-12 • 4.2 | 4.2 | 0.7 | 0.7 | 30 | 5 | 5.1 | 3 | 0.8 | 0.9 | 60 | 7

Core • 13-22 • 0.06 | 8.5 | 0.007 | 12.5 | 0.9 | 0.04 | 9.2 | 0.003 | 22.5 | 0.6

Challenge • 23-26 • 0.0405 | 0.005 | 0.0302 | 0.008

2.3 • Exponent Shorthand

OUT OF 18

Warm-Up • 1-8 • 10 | 100 | 1000 | 10000 | 5 | 1000000 | 6 | 100000

Core • 9-18 • 300 | 4500 | 70000 | 600 | 250 | 600 | 3500 | 90000 | 800 | 150

Challenge • 19-22 • 100000 | 350 | 1000000 | 450

2.4 • Patterns in Zeros

OUT OF 22

Warm-Up • 1-12 • 400 | 40 | 4000 | 4 | 3 | 400 | 600 | 60 | 6000 | 6 | 600 | 3

Core • 13-22 • 2500 | 25 | 2.5 | 1500 | 15 | 3500 | 35 | 3.5 | 2500 | 25

Challenge • 23-26 • 25 | 100 | 45 | 100

Fri • Zoom Atlas Begins

OUT OF 7

Warm-Up • 1-3 • 30 | 300 | 50

Core • 4-7 • 4 | 25 | 6 | 48

Challenge • 8-13 • 4 | 25 | 10000 | 7 | 35 | 10000

Round to a Named Place

Find the place, look one to its right, decide.

Warm-Up 11 ITEMS • RECALL

1 Round 47 to the nearest ten

2 Round 43 to the nearest ten

3 Round 250 to the nearest hundred

4 Round 4.7 to the nearest whole

5 Round 3.2 to the nearest whole

6 Round 9.8 to the nearest whole

7 Round 830 to the nearest hundred

8 Round 6.3 to the nearest whole

9 Round 5.5 to the nearest whole

10 Round 1,240 to the nearest thousand

11 Round 96 to the nearest ten

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

12. Round 4,829 to the nearest thousand

13. Round 4,829 to the nearest hundred

14. Round 12.5 to the nearest whole

15. Round 0.48 to the nearest tenth

16. Round 2.451 to the nearest hundredth

17. Round 4,672 to the nearest hundred

18. Round 28,499 to the nearest thousand

19. Round 1,950 to the nearest hundred

20. Round 74,500 to the nearest thousand

21. Round 349 to the nearest hundred

Rounding Decimals

Same method, smaller places.

Warm-Up 12 ITEMS • RECALL

1 Round 0.4 to the nearest whole

2 Round 0.6 to the nearest whole

3 Round 1.5 to the nearest whole

4 Round 2.4 to the nearest whole

5 Round 0.35 to the nearest tenth

6 Round 0.24 to the nearest tenth

7 Round 2.7 to the nearest whole

8 Round 0.62 to the nearest tenth

9 Round 9.5 to the nearest whole

10 Round 0.38 to the nearest tenth

11 Round 4.04 to the nearest tenth

12 Round 0.95 to the nearest tenth

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Round 3.456 to the nearest hundredth

14. Round 3.456 to the nearest tenth

15. Round 0.075 to the nearest hundredth

16. Round 12.349 to the nearest tenth

17. Round 7.005 to the nearest hundredth

18. Round 5.628 to the nearest hundredth

19. Round 0.4449 to the nearest hundredth

20. Round 12.996 to the nearest hundredth

21. Round 3.0451 to the nearest thousandth

22. Round 7.185 to the nearest hundredth

Challenge

23. Round 0.9999 to the nearest thousandth

24. Round 2.4999 to the nearest whole

25. Round 0.99951 to the nearest thousandth

26. The smallest number rounding to 2.5 at the nearest tenth



Working space



Error journal

Which Place Matters

Money rounds to hundredths. Distance rarely does.

Warm-Up 11 ITEMS • RECALL

1 Round \$4.56 to the nearest cent

2 Round \$4.567 to the nearest cent

3 Round 4.7 to the nearest whole

4 Round 250 to the nearest hundred

5 Round 0.5 to the nearest whole

6 Round \$9.49 to the nearest dollar

7 Money is usually rounded to the nearest — type cent or dollar

8 Round 4.5 km to the nearest km

9 Round 18 minutes to the nearest ten minutes

10 Round \$12.49 to the nearest dollar

11 Round 2.2 kg to the nearest kg

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

12. Round \$19.99 to the nearest dollar

.....

13. Round 1.005 m to the nearest centimetre

.....

14. Round 47 minutes to the nearest ten

.....

15. 4.4 buses of people — buses needed

.....

16. Round 2.44 km to the nearest tenth

.....

17. A crowd of 48,720 to the nearest thousand

.....

18. That crowd to the nearest ten thousand

.....

19. A 3.456 second time to the nearest hundredth

.....

20. A \$1,499 price to the nearest hundred dollars

.....

21. 12.4 cm to the nearest cm

.....



Working space



Error journal

Estimate to Check

Round first, compute second, compare.

Warm-Up 12 ITEMS • RECALL

1 Estimate $4.9 + 3.1$ 2 Estimate $9.8 - 4.9$ 3 Estimate 2.1×4 4 Estimate $11.9 \div 4$ 5 Estimate $0.9 + 1.1$ 6 Estimate 5.2×2 7 Estimate $41 + 38$ 8 Estimate $6.1 + 2.9$ 9 Estimate 19×5 10 Estimate $402 - 199$ 11 Estimate $9.8 + 0.3$ 12 Estimate 51×2

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. Estimate 412×19 14. Estimate 0.49×100 15. True value of 0.49×100 16. Estimate $19.8 \div 5$ 17. Estimate $12.4 - 7.9$ 18. Estimate $4.87 + 3.14$ 19. True value of $4.87 + 3.14$ 20. Estimate 19.6×5 21. True value of 19.6×5 22. Estimate $6,120 - 2,890$



Mid-Unit Quiz

Eight items across Weeks 1–3. 85% to keep flying.

✦ Warm-Up 4 ITEMS • RECALL

1 The 7 in 7,000 is how many times the 7 in 700

2 Write four hundred six thousandths

3 4.5×10

4 Round 3.6 to the nearest whole

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.

5. 0.06×100

6. $850 \div 10^2$

7. Write 10^4 as an ordinary number

8. Round 2.451 to the nearest hundredth

9. Larger: 0.406 or 0.41

10. 0.07×1000

11. Round 8.472 to the nearest hundredth

12. $920 \div 100$

13. 6×10^3

14. Write three and six hundredths

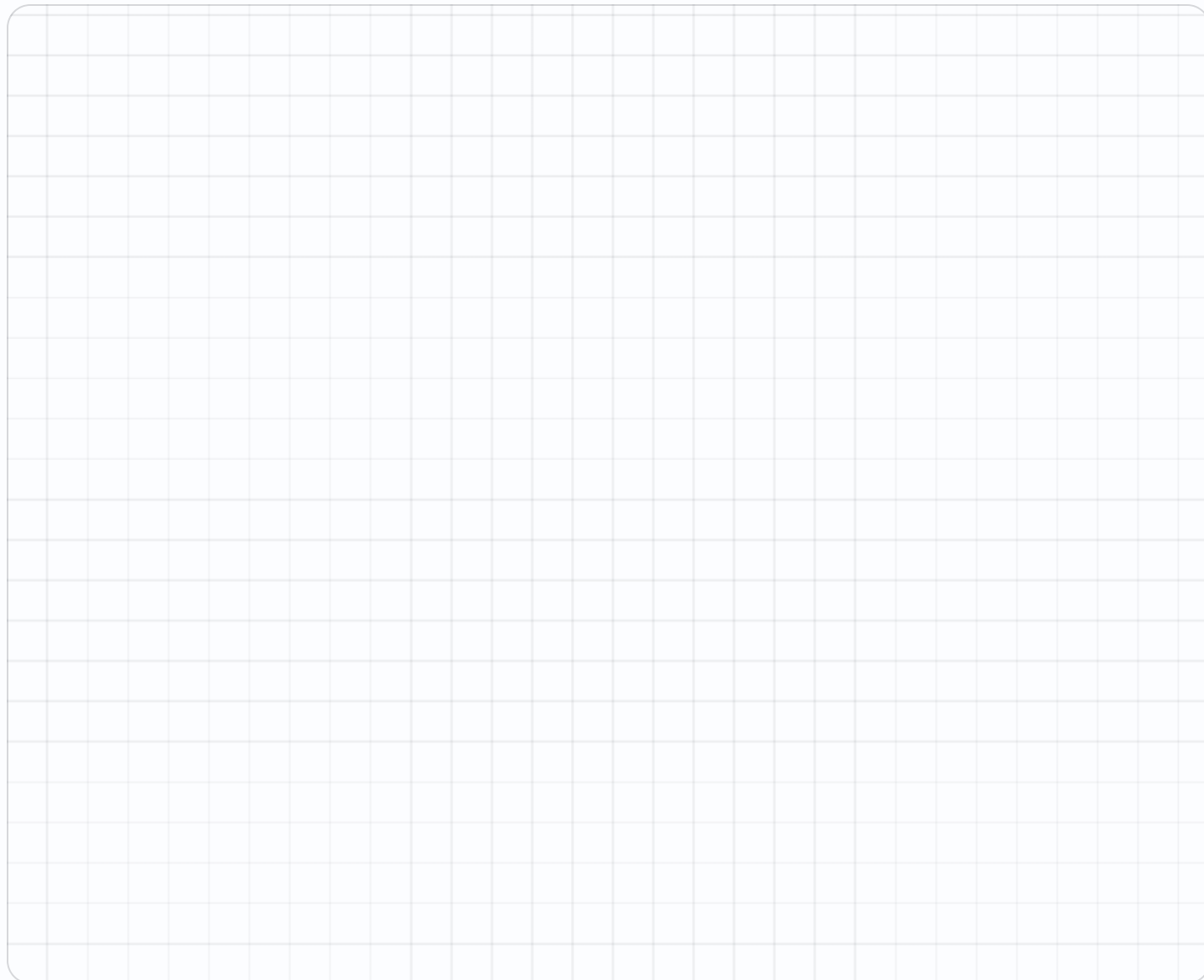
★ Challenge NOT GRADED • REASON IT OUT

15. Which power of ten takes 0.7 to 7,000

.....

16. How many times bigger is 700 than 0.07

.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Week 3 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

3.1 • Round to a Named Place

OUT OF 21

Warm-Up • 1-11 • 50 | 40 | 300 | 5 | 3 | 10 | 800 | 6 | 6 | 1000 | 100

Core • 12-21 • 5000 | 4800 | 13 | 0.5 | 2.45 | 4700 | 28000 | 2000 | 75000 | 300

Challenge • 22-25 • 10 | 0.048 | 8450000 | 449

3.2 • Rounding Decimals

OUT OF 22

Warm-Up • 1-12 • 0 | 1 | 2 | 2 | 0.4 | 0.2 | 3 | 0.6 | 10 | 0.4 | 4 | 1

Core • 13-22 • 3.46 | 3.5 | 0.08 | 12.3 | 7.01 | 5.63 | 0.44 | 13 | 3.045 | 7.19

Challenge • 23-26 • 1 | 2 | 1 | 2.45

3.3 • Which Place Matters

OUT OF 21

Warm-Up • 1-11 • 4.56 | 4.57 | 5 | 300 | 1 | 9 | cent | 5 | 20 | 12 | 2

Core • 12-21 • 20 | 1.01 | 50 | 5 | 2.4 | 49000 | 50000 | 3.46 | 1500 | 12

Challenge • 22-25 • 0.08 | 2.99 | 0.5 | second

3.4 • Estimate to Check

OUT OF 22

Warm-Up • 1-12 • 8 | 5 | 8 | 3 | 2 | 10 | 80 | 9 | 100 | 200 | 10 | 100

Core • 13-22 • 8000 | 50 | 49 | 4 | 4 | 8 | 8.01 | 100 | 98 | 3000

Challenge • 23-26 • 250000 | 4.544 | 41 | 300

Fri • Mid-Unit Quiz

OUT OF 14

Warm-Up • 1-4 • 10 | 0.406 | 45 | 4

Core • 5-14 • 6 | 8.5 | 10000 | 2.45 | 0.41 | 70 | 8.47 | 9.2 | 6000 | 3.06

Challenge • 15-16 • 10000 | 10000

Finish the Atlas

Every jump labelled with its power of ten.

Warm-Up 12 ITEMS • RECALL

1 A 3 cm object at 10 times — cm

2 At 100 times — cm

3 A 4 m object at one hundredth — cm

4 A 250 cm object at one tenth — cm

5 0.5 mm at 10 times — mm

6 A 2.5 cm object at 100 times — cm

7 A 2 cm object at 10 times — cm

8 At 1000 times — cm

9 A 500 cm object at one tenth — cm

10 A 8 m object at one hundredth — cm

11 $10 \times 10 \times 10$

12 A 4 mm object at 100 times — mm

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. A 4,000 mm object at one thousandth — mm

.....

14. A 2.5 cm object at 1000 times — metres

.....

15. From 0.5 mm to 5 m — the multiplier

.....

16. 3.5 m at 100 times — metres

.....

17. 0.035×10^4

.....

18. A 1.5 cm bug drawn 20 times bigger — cm

.....

19. A 12 m room drawn at one hundredth — cm

.....

20. A 0.5 mm grain at 1000 times — mm

.....

21. From 3 cm to 3 m — the multiplier

.....

22. From 4 km to 4 cm — the divisor

.....



Working space



Error journal

Explain a Jump

Pick one panel and say what moved and what stayed.

Warm-Up 12 ITEMS • RECALL

1 4.2×10

2 $42 \div 10$

3 0.7×100

4 $70 \div 100$

5 3×1000

6 $3000 \div 1000$

7 0.5×10

8 5×10

9 $50 \div 10$

10 $5 \div 10$

11 0.05×100

12 $500 \div 100$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. 0.06×100

14. $6 \div 100$

15. 2.4×1000

16. $2400 \div 1000$

17. 0.008×1000

18. Jumping from 0.05 to 500 — the multiplier

19. Jumping from 4000 to 0.4 — the divisor

20. How many tenths in 3

21. How many hundredths in 0.5

22. How many thousandths in 0.02



Working space



Error journal

Mixed Review

Place value, powers and rounding in one set.

Warm-Up 12 ITEMS • RECALL

1 The 4 in 400 is how many times the 4 in 40

2 One tenth of 500

3 10^3

4 Zeros in 10^5

5 Larger: 0.5 or 0.35

6 Round 4.7 to the nearest whole

7 $0.4 + 0.5$

8 $1 - 0.3$

9 3×100

10 Round 7.6 to the nearest whole

11 0.2×10

12 $80 \div 10$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. 3.5×100

.....

14. $42 \div 100$

.....

15. 0.035×100

.....

16. Round 4,829 to the nearest thousand

.....

17. Smallest of 0.6, 0.55, 0.506, 0.65

.....

18. $4.5 + 0.75$

.....

19. 0.06×1000

.....

20. Round 5.349 to the nearest hundredth

.....

21. $9 \div 100$

.....

22. 7×10^4

.....



Working space



Error journal

Error Journal Sweep

Re-read every entry from the mission. Fix only what repeats.

Warm-Up 12 ITEMS • RECALL

1 Tenths in 0.3

2 10^2

3 4.2×10

4 $7 \div 1000$

5 Is 0.4 the same as 0.40 — yes or no

6 0.05 — name the last place

7 Round 4.5 to the nearest whole

8 0.6×100

9 Zeros in 10^4

10 Write two tenths

11 $70 \div 100$

12 Bigger — type 0.8 or 0.79

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. $0.125 + 0.875$

.....

14. $850 \div 100$

.....

15. 0.06×100

.....

16. Round 9.96 to the nearest tenth

.....

17. Larger: 0.08 or 0.075

.....

18. Round 0.0649 to the nearest hundredth

.....

19. 0.035×100

.....

20. Write four and twelve thousandths

.....

21. 3.5×10^3

.....

22. Bigger — type 0.409 or 0.41

.....



Working space



Error journal

Mission 01 Test

Twelve items plus the Big Question, answered out loud.

Warm-Up 4 ITEMS • RECALL

1 The 3 in 300 is how many times the 3 in 3
_____2 Write sixty-two thousandths
_____3 5×10
_____4 Round 2.4 to the nearest whole

Core GRADED • SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.5. Hundredths in 0.3
.....6. 2.4×10^3
.....7. $7 \div 1000$
.....8. Round 9.96 to the nearest tenth
.....9. Round 0.0475 to the nearest thousandth
.....10. Which power of ten takes 0.7 to 700
.....11. Smallest of 0.6, 0.55, 0.506, 0.65
.....12. 0.035×10^4
.....13. 0.08×1000
.....14. $640 \div 100$
.....15. Round 7.462 to the nearest tenth
.....16. 4×10^3
.....17. Write eight and five hundredths
.....18. Bigger — type 0.51 or 0.509
.....19. How many times bigger is 300 than 30
.....20. Round 15.849 to the nearest hundredth
.....



Week 4 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

4.1 • Finish the Atlas

OUT OF 22

Warm-Up • 1-12 • 30 | 300 | 4 | 25 | 5 | 250 | 20 | 2000 | 50 | 8 | 1000 | 400

Core • 13-22 • 4 | 25 | 10000 | 350 | 350 | 30 | 12 | 500 | 100 | 100000

Challenge • 23-26 • 40.5 | 4 | 10000000 | 2000

4.2 • Explain a Jump

OUT OF 22

Warm-Up • 1-12 • 42 | 4.2 | 70 | 0.7 | 3000 | 3 | 5 | 50 | 5 | 0.5 | 5 | 5

Core • 13-22 • 6 | 0.06 | 2400 | 2.4 | 8 | 10000 | 10000 | 30 | 50 | 20

Challenge • 23-26 • 0.0405 | 100 | 750 | 1000

4.3 • Mixed Review

OUT OF 22

Warm-Up • 1-12 • 10 | 50 | 1000 | 5 | 0.5 | 5 | 0.9 | 0.7 | 300 | 8 | 2 | 8

Core • 13-22 • 350 | 0.42 | 3.5 | 5000 | 0.506 | 5.25 | 60 | 5.35 | 0.09 | 70000

Challenge • 23-26 • 25 | 2.45 | 6.004 | 10000

Thu • Error Journal Sweep

OUT OF 22

Warm-Up • 1-12 • 3 | 100 | 42 | 0.007 | yes | hundredths | 5 | 60 | 4 | 0.2 | 0.7
| 0.8

Core • 13-22 • 1 | 8.5 | 6 | 10 | 0.08 | 0.06 | 3.5 | 4.012 | 3500 | 0.41

Challenge • 23-26 • 4.544 | 40.5 | 10 | 100000

Fri • Mission 01 Test

OUT OF 20

Warm-Up • 1-4 • 100 | 0.062 | 50 | 2

Core • 5-20 • 30 | 2400 | 0.007 | 10 | 0.048 | 1000 | 0.506 | 350 | 80 | 6.4 | 7.5
| 4000 | 8.05 | 0.51 | 10 | 15.85

Challenge • 21-24 • 3.407 | 4 | 25 | 420